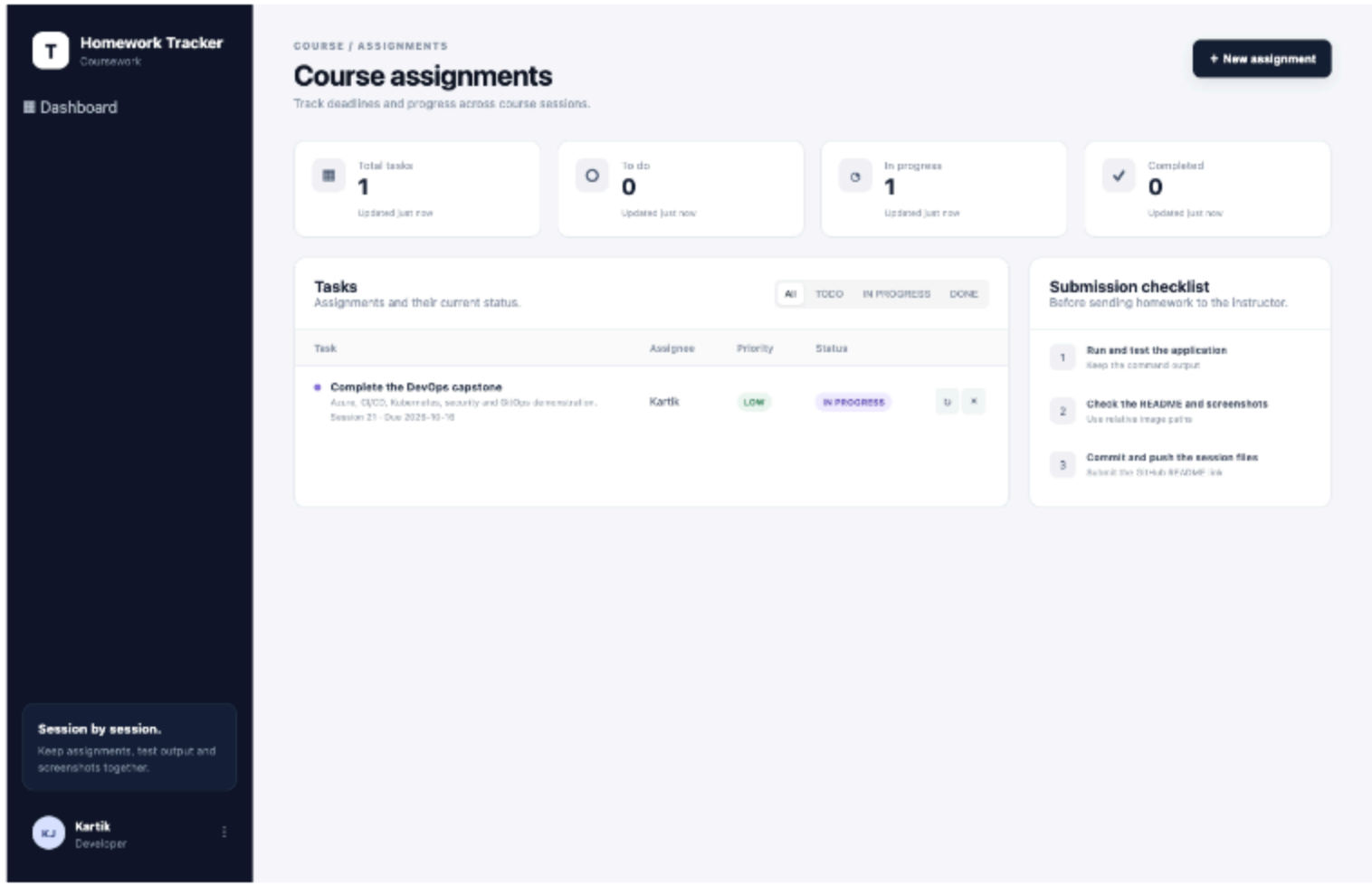
DEVOPS FINAL CAPSTONE

Homework Tracker

Kartik Jangid

React, FastAPI and PostgreSQL
Azure deployment with K3s

The application



Application after the Azure upgrade, October 2026

Assignments for Sessions 1–21

Deadline, assignee and priority

Create, edit, filter and track status

Application and delivery architecture

Release delivery



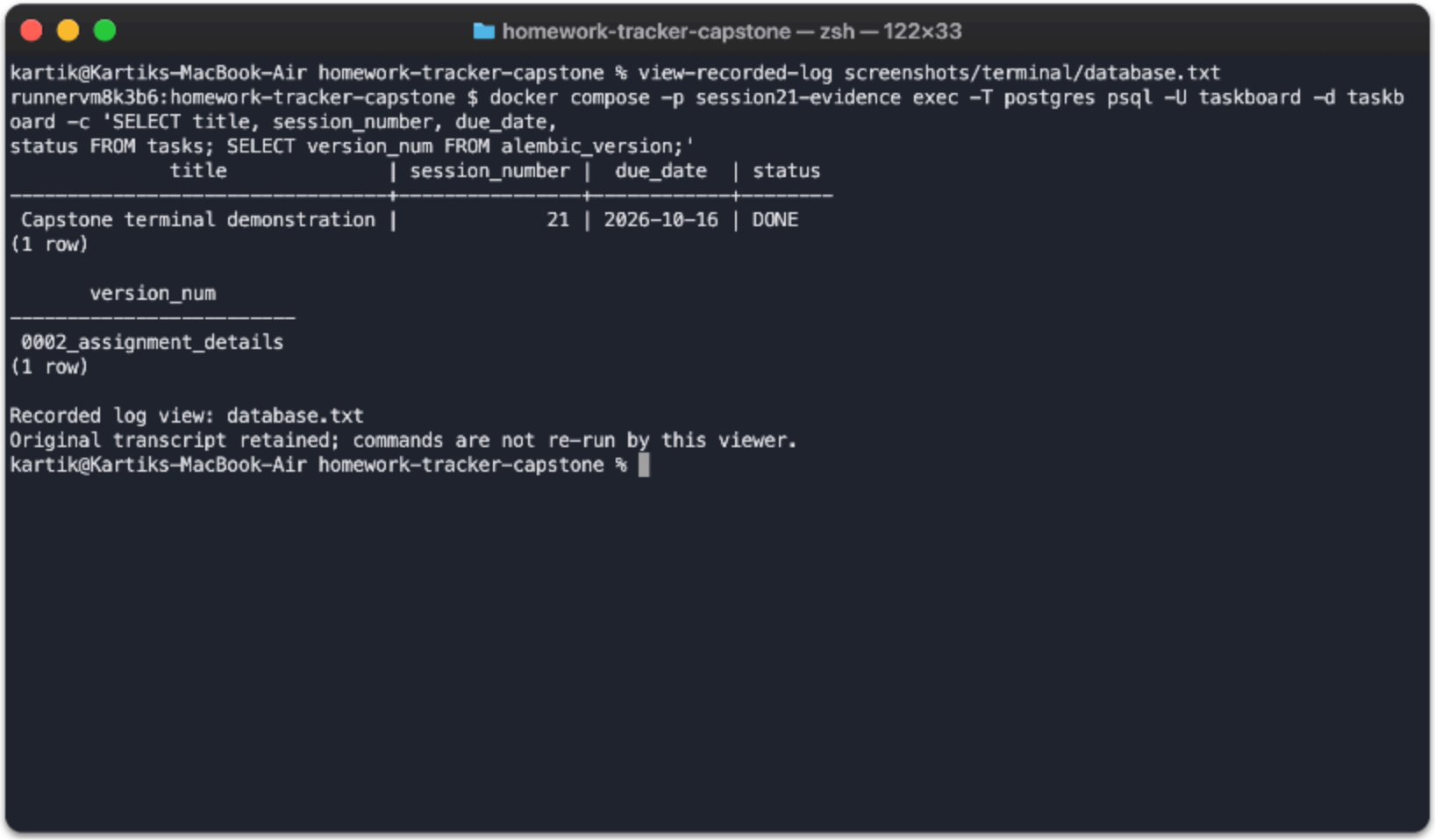
Application requests



Terraform provisions the Azure VM and network. CI verifies the chart in Kind.

Prometheus scrapes FastAPI metrics. Grafana displays the results.

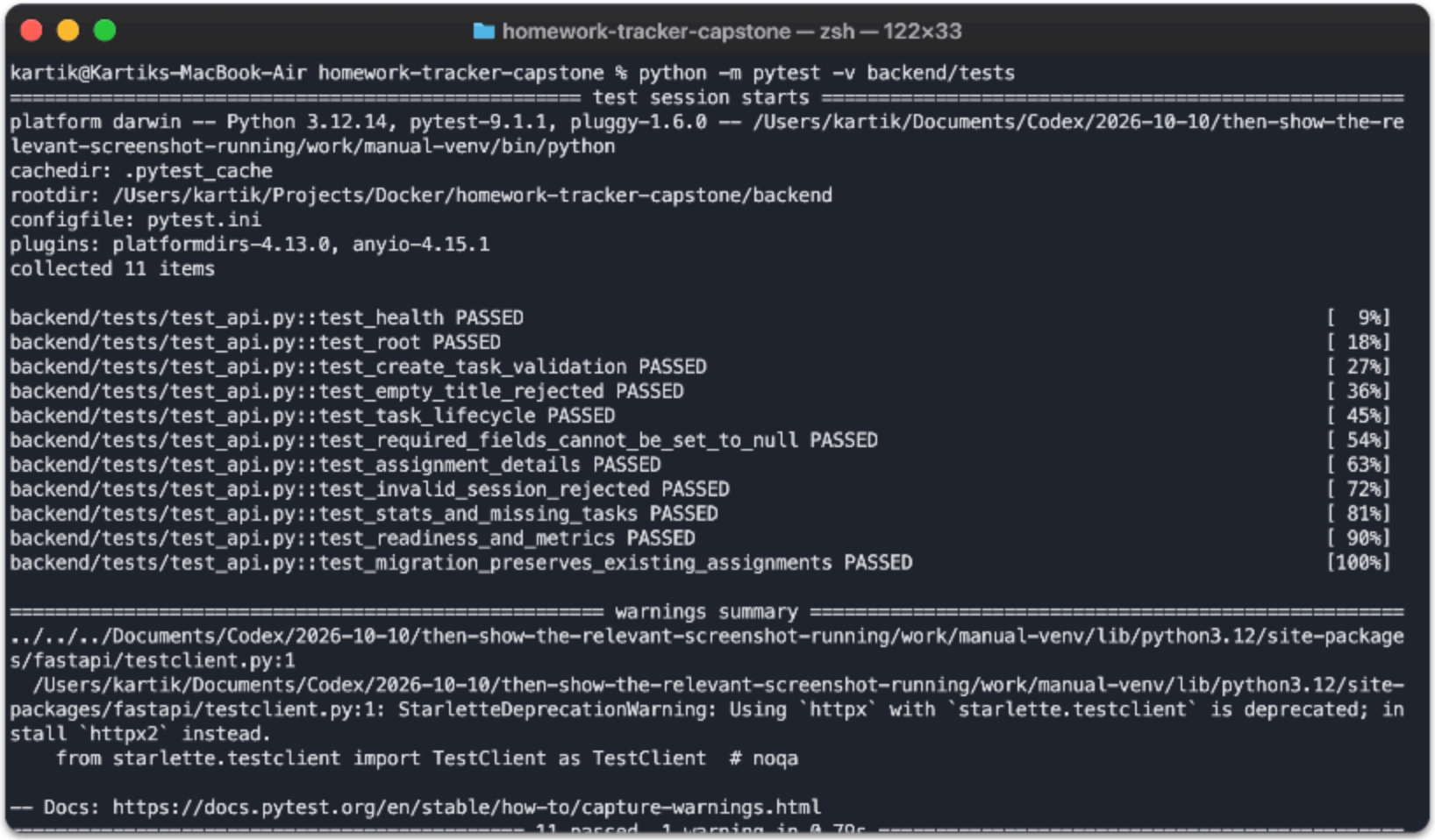
API and database



Recorded database output viewed in Mac zsh

- CRUD endpoints manage assignments
- Swagger documents the API
- Alembic versions database changes
- PostgreSQL retained data through the upgrade

Automated tests



Local backend test rerun, 17 October 2026

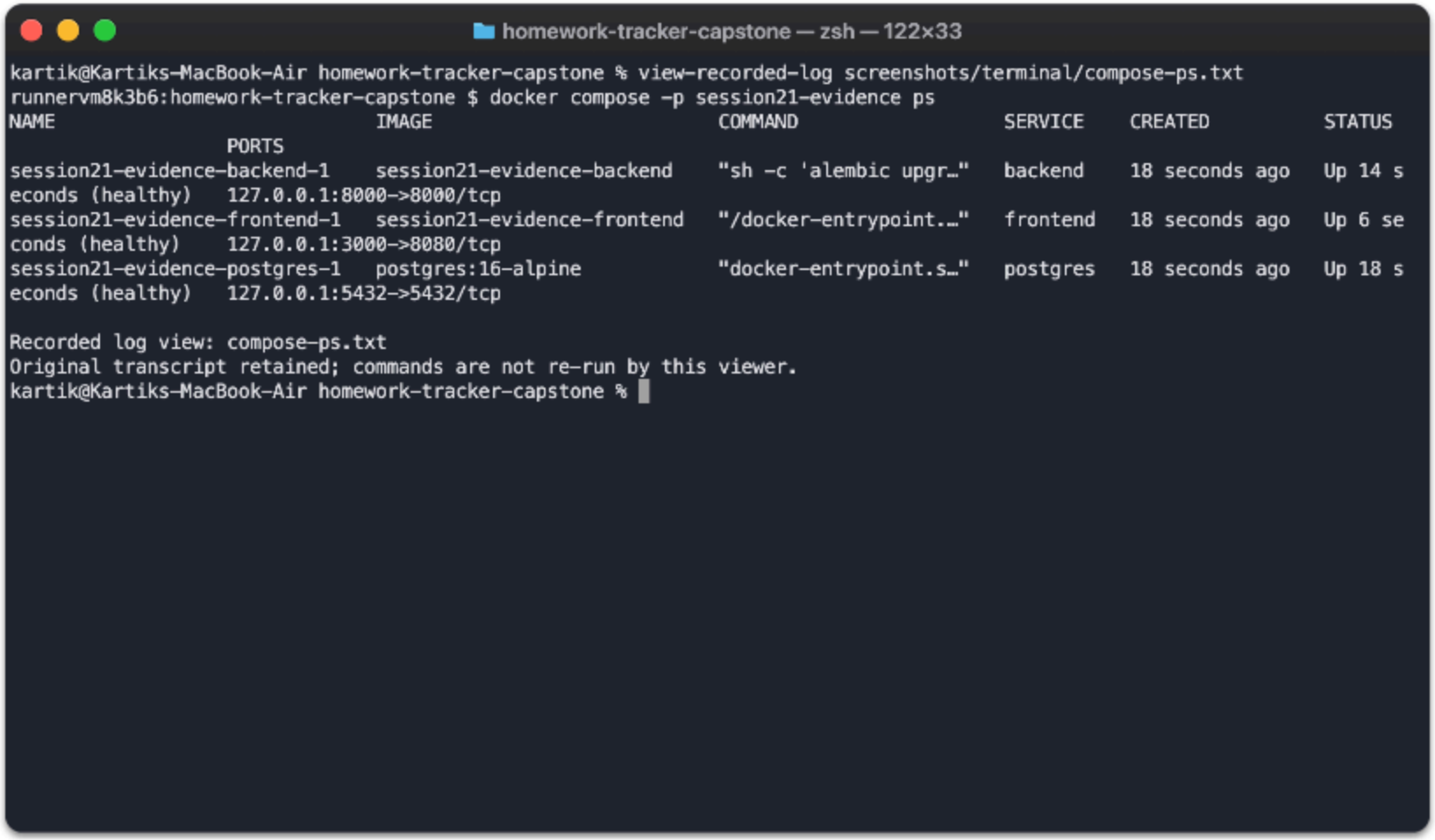
11 API and migration tests passed

Tests use an isolated test database

Failures stop the pipeline

The run reported one deprecation warning

Docker and Compose



Recorded clean Compose verification viewed in Mac zsh

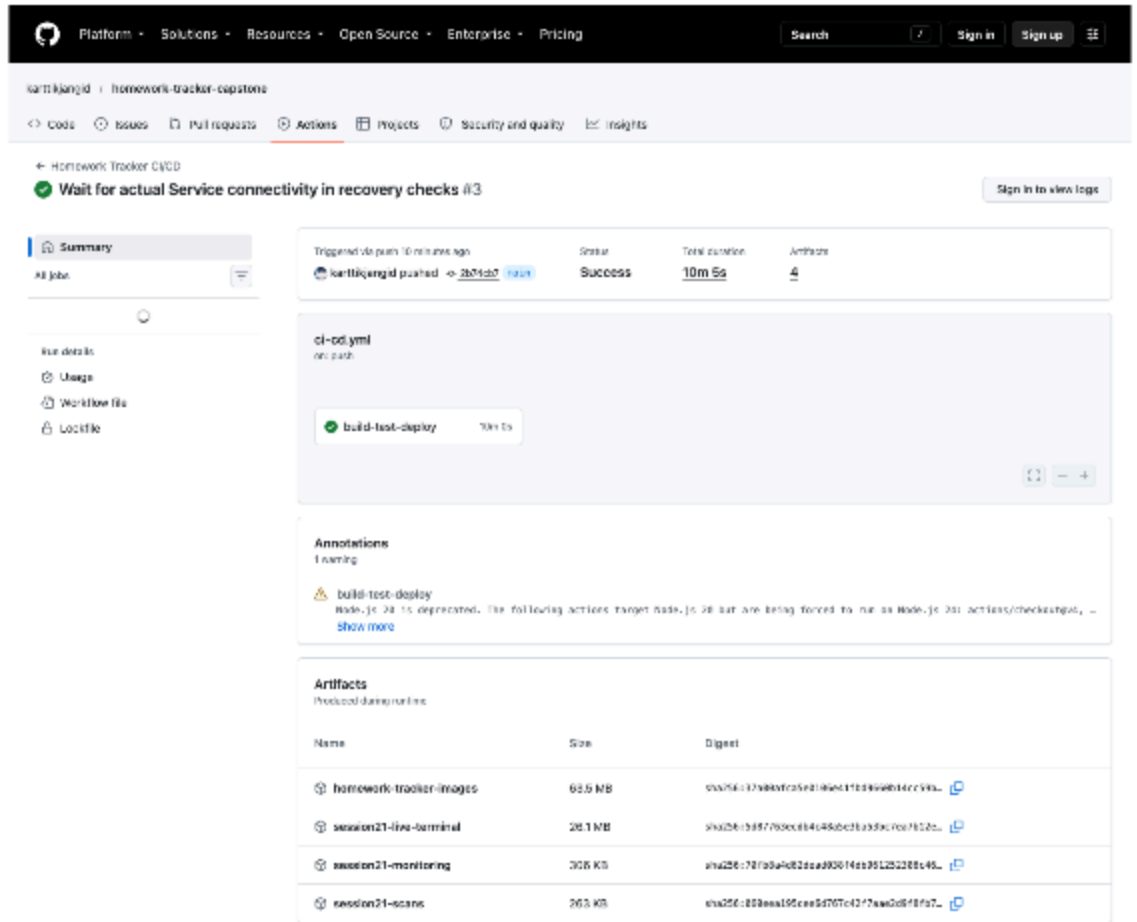
Frontend, backend and PostgreSQL

Multi-stage frontend image

Both application images run as non-root

Health checks verify readiness

GitHub Actions and image delivery



Tests and frontend build

Security gates before publication

Two images published to GHCR

Commit SHA identifies the image release

Successful application pipeline, run 38812142701

Security gates

Bandit checks Python source

pip-audit and npm audit inspect dependencies

Trivy scans the repository for secrets

Trivy scans both container images

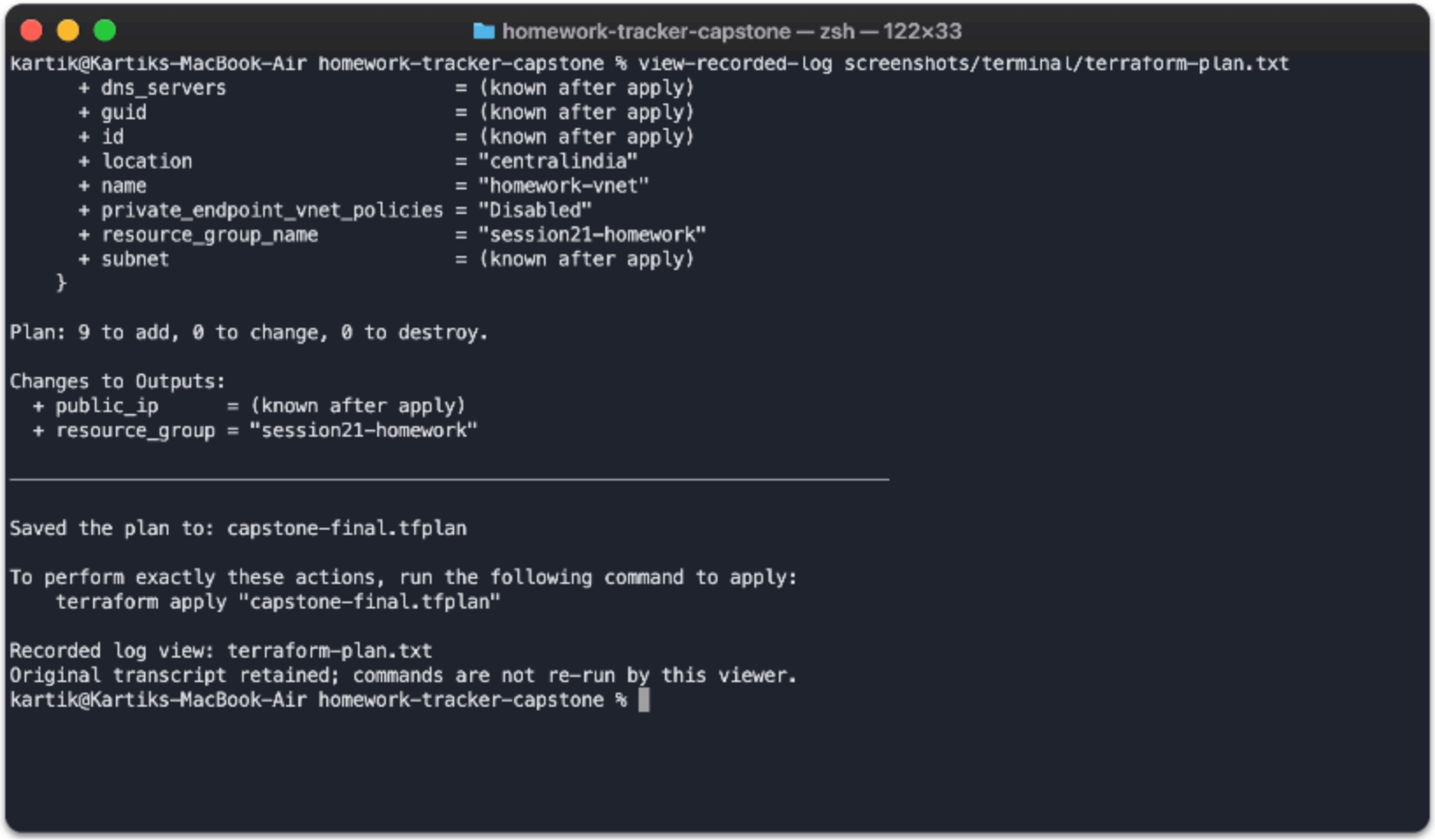
HIGH or CRITICAL image findings fail the build

The saved release scans passed

A clean result covers that scan and dependency set

New dependency or image changes require another scan

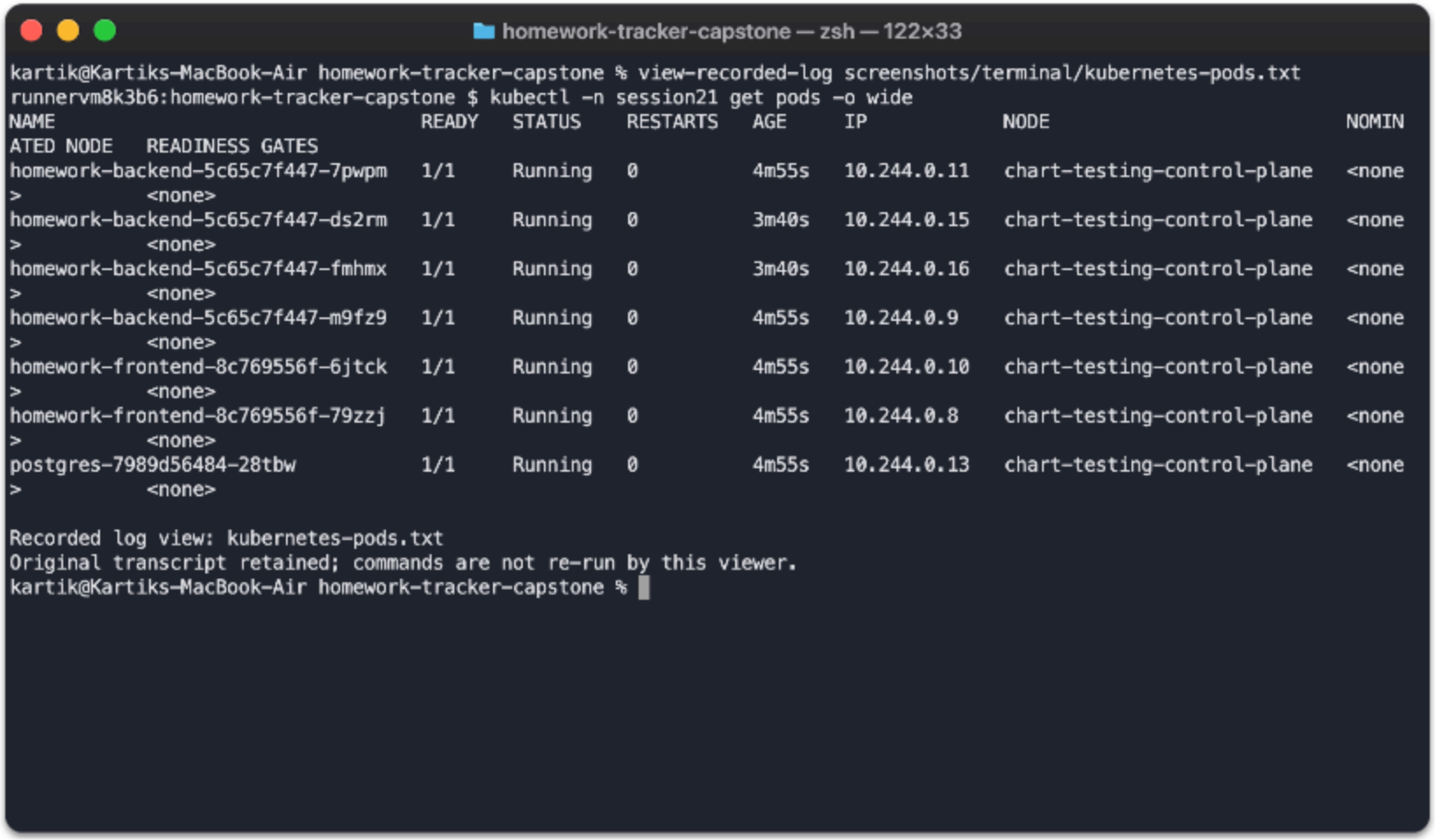
Terraform on Azure



Recorded Terraform plan viewed in Mac zsh

- VNet, subnet and restricted NSG
- One Standard_B2als_v2 VM
- K3s runs the Kubernetes workload
- Kubernetes API access uses an SSH tunnel

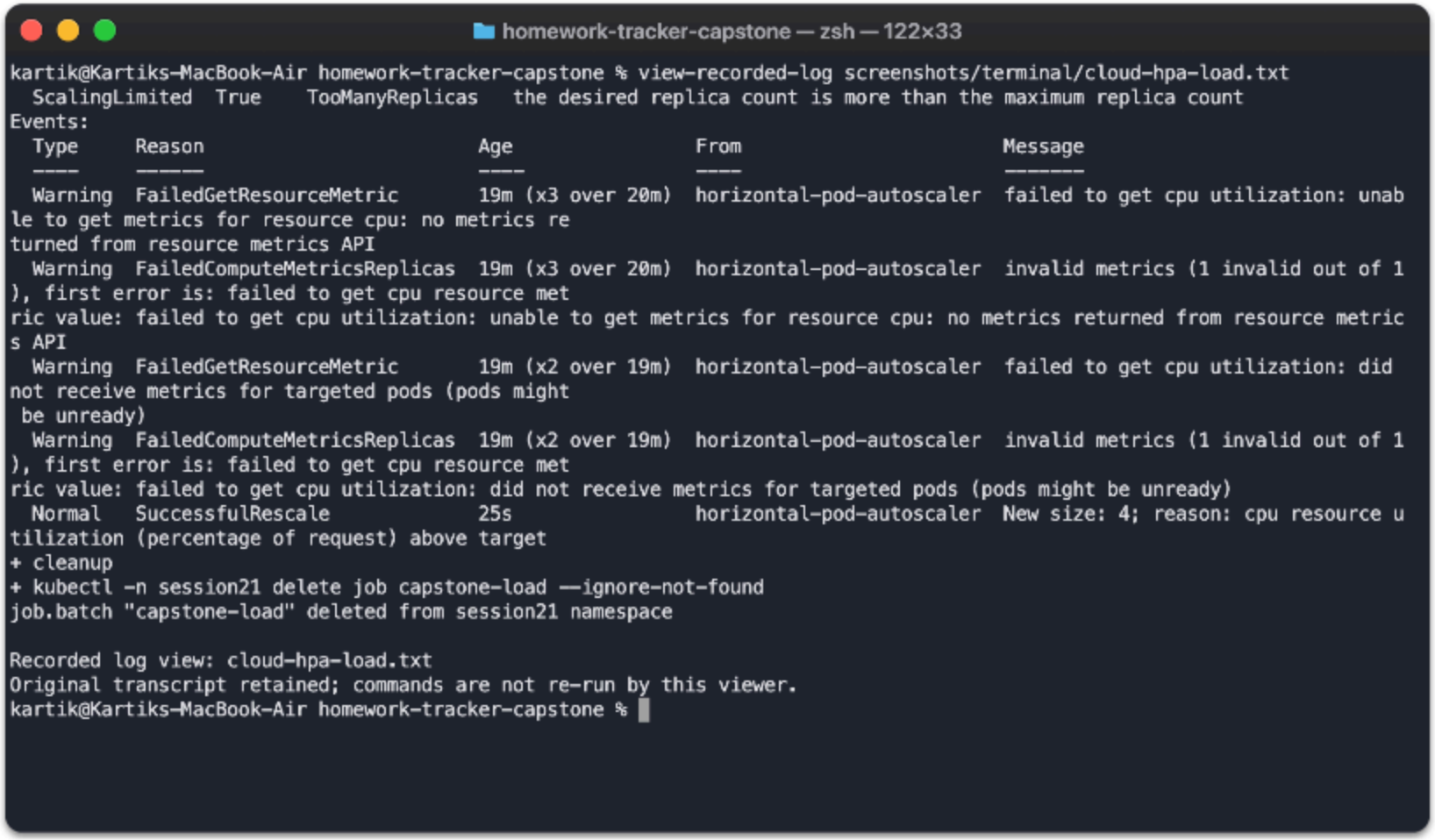
Kubernetes and Helm



Recorded Pod status viewed in Mac zsh

- Frontend and backend Deployments
- ClusterIP Services and Ingress
- ConfigMap, Secret and readiness probes
- PostgreSQL PVC keeps application data

Autoscaling under load



Recorded Azure HPA results viewed in Mac zsh

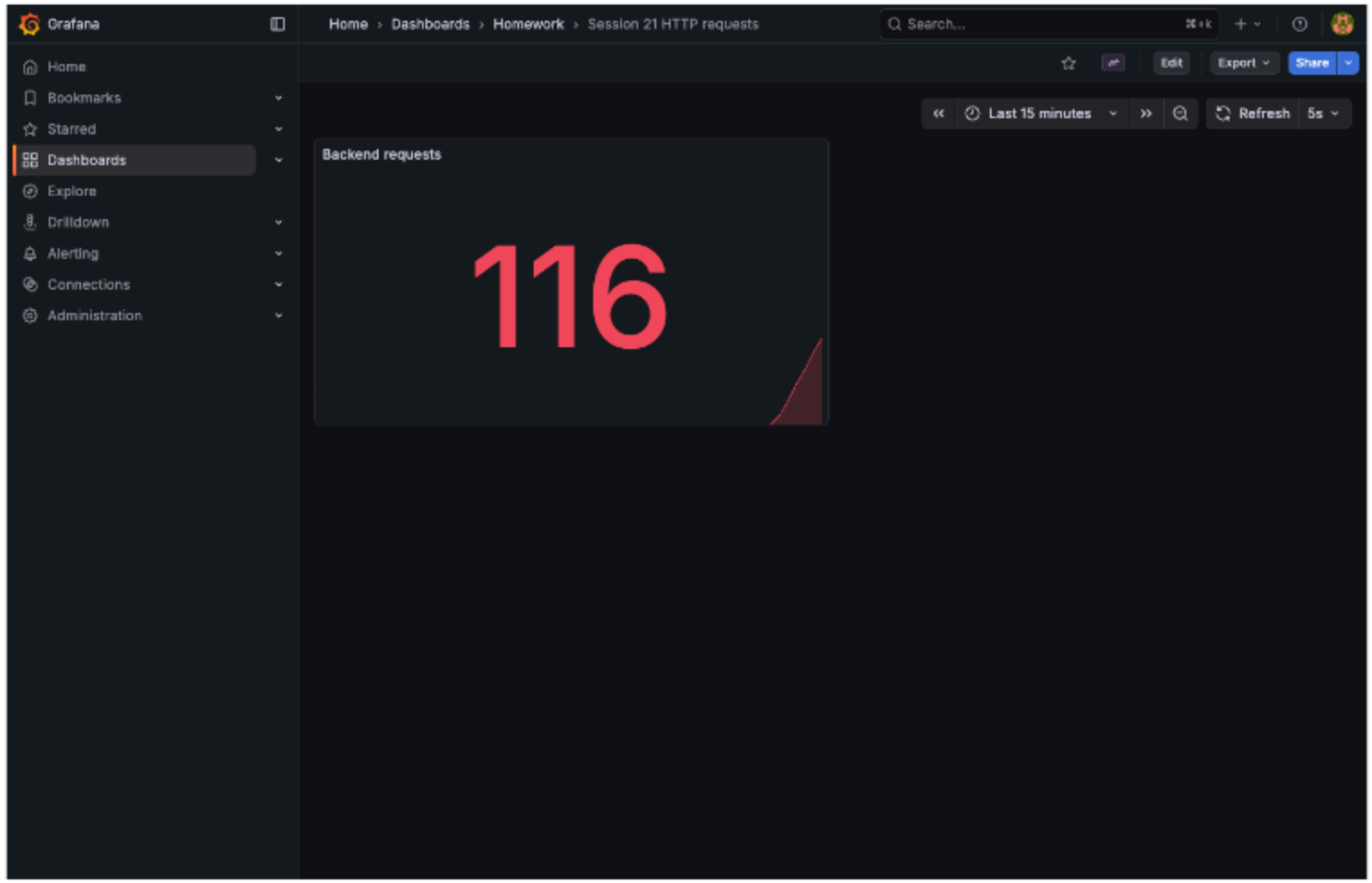
HTTP load increases backend CPU use

HPA observes CPU metrics

Backend scaled from 2 to 4 Pods

The load test uses bounded duration

Prometheus and Grafana



Grafana dashboard during the Azure demonstration

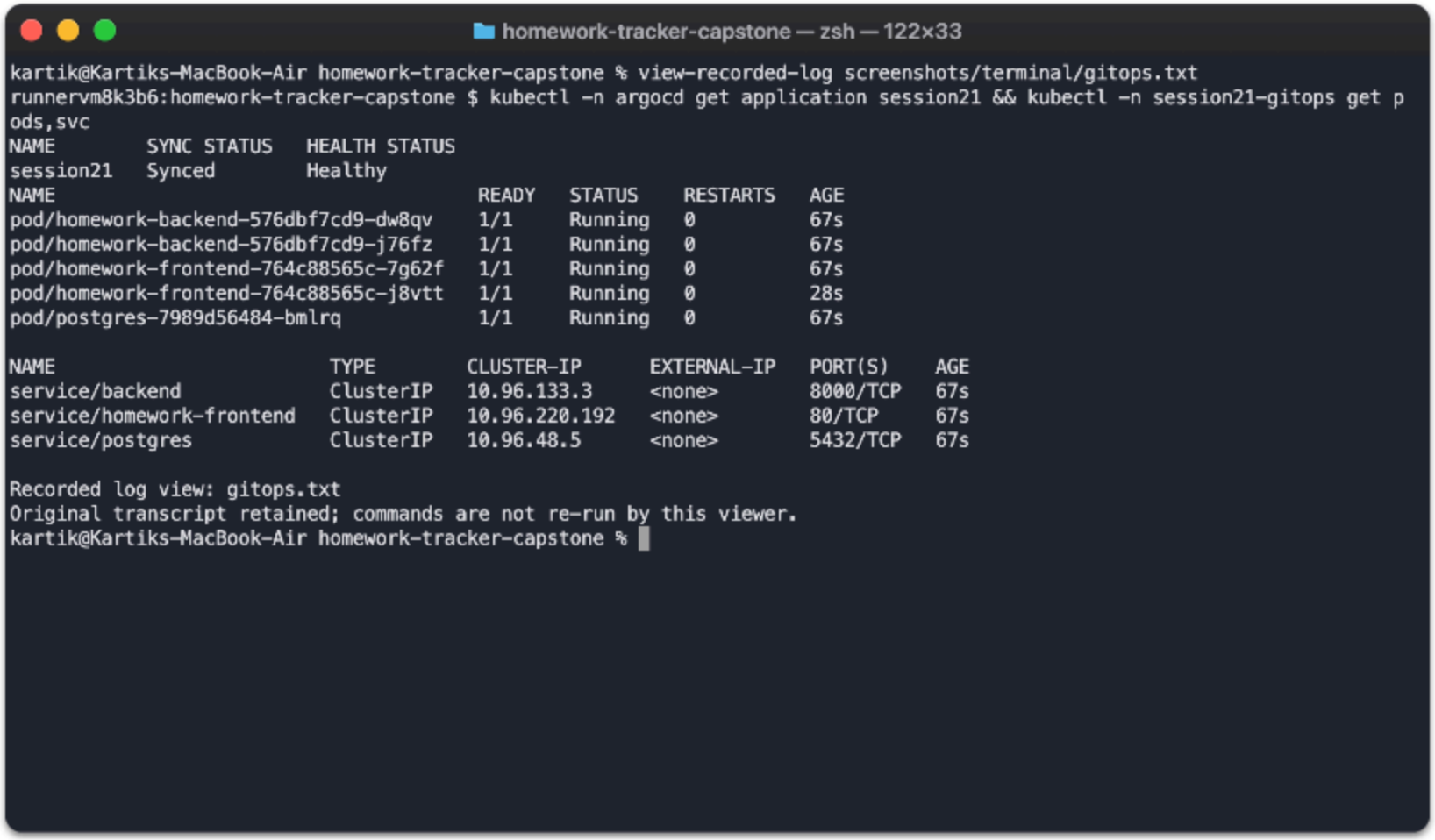
FastAPI exposes /metrics

Prometheus scrapes the backend

The target reported UP

Grafana shows application request metrics

GitOps reconciliation



Recorded isolated CI GitOps verification viewed in Mac zsh

Git stores the declared configuration

Argo CD compares desired and live state

The CI exercise changes a replica count

Self-healing restores Synced and Healthy

Failure investigation and recovery

Invalid image tag: inspect Pod events, fix the tag, verify rollout

Wrong Service selector: inspect labels and EndpointSlices, restore the selector, test HTTP

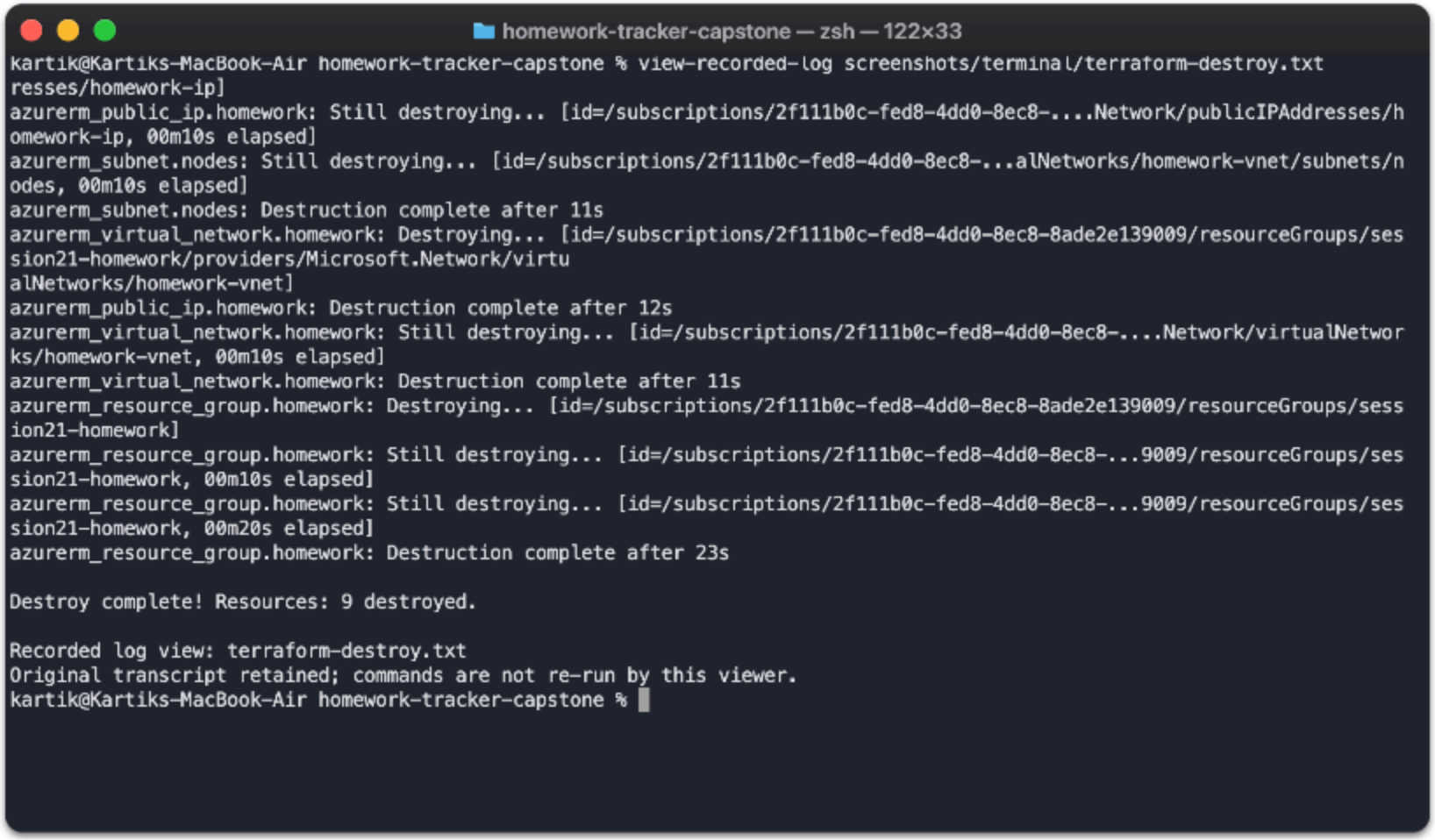
Early HTTP probe: retry for a bounded period while requiring a successful response

A first standalone recovery check failed

The corrected check passed in the successful run

Both the failure and recovery records remain available

Cleanup and project limits



Recorded Terraform destroy viewed in Mac zsh

Terraform destroyed all 9 cloud resources

The saved group-existence check returned false

The Azure demo endpoint is no longer hosted

Authentication remains outside this classroom demo

Key learnings

Database readiness must come before application traffic

Pod readiness alone does not prove Service connectivity

SHA tags identify exactly which images were deployed

Inspect raw test failures and preserve recovery evidence

Destroy temporary cloud infrastructure after verification

Presentation demo and references

Create an assignment and show the API and database

Commit a small change, push and follow the pipeline

Verify the image SHA and deploy the approved release

Check Ingress, autoscaling and monitoring

Explain a failure, recovery and cleanup

Repository and source code

github.com/karttikjangid/homework-tracker-capstone

Recorded walkthrough: docs/walkthrough.md

Instructor brief and rubric: docs/course/